

Basics of Agriculture

HuNu2073



**BETTER
PRODUCTION**



**BETTER
NUTRITION**



**BETTER
ENVIRONMENT**



**BETTER
LIFE**



1. NATURAL RESOURCES MANAGEMENT

“Caring for Earth is caring for our self and our community” Daniel Christian Wahl

“Healing the land is healing ourselves.” Adrianna Acosta

1.1. Introduction to Natural Resources

- A resource is a form of energy and or matter which is essential for the functioning of living organisms (population and ecosystem). In other words, any part of the environment such as land, water, air, minerals, forest, wild life etc. that the man can utilize to promote his welfare may be regarded as resources.
- A natural capital /resource/ is any asset that can be obtained from the environment e.g. water, soil, plants, wind, animals, minerals, the energy of the sun and many others.

1.1.1. Classification of Natural Resources

A prominent way to classify natural resources is to look in terms of depletion risk.

- A renewable natural resource is one that can regrow, or whose supplies can be replenished through natural processes. E.g. plants, animals, insects, or wind.
- A non-renewable natural resource is one that can be used up or that will not replenish itself. E.g. oil, coal, minerals, or rocks. The use of these resources should be carefully monitored and managed according to their availability. The effects that their exploitation has on all the other resources and the environment as a whole should be carefully followed.

1.1.2. Ecosystem

What is an ecosystem?

What is an Ecosystem Structure?

The **structure** of an **ecosystem** is essentially a description of an environment's organisms and physical features.

It is described under two major headings: Abiotic components and Biotic components.

Ecosystem services

- Ecosystem services are the direct and indirect contributions of ecosystems to human well-being. Healthy, bio-diverse ecosystems sustain humanity by providing air, water and other essential services.
- Many (ecosystem) goods and services e.g. wildlife habitat and diversity, watershed services, carbon storage, and scenic landscapes - are traditionally viewed as free benefits to society, or 'public goods'. Lacking a formal market, these natural assets are traditionally absent from society's balance sheet; their critical contributions are often overlooked in public, corporate, and individual decision-making.
- The degradation of ecosystems has major consequences for our own health and wellbeing. Healthy ecosystems provide us with a wealth of ecosystem goods and services such as water purification, crop pollination, flood protection, **carbon storage and sequestration**. Their monetary value is considerable and virtually irreplaceable.

...Ecosystem services



1.1.3. Over-exploitation

- Being renewable does not mean that these resources automatically last forever. If the renewable resources in a particular area are overexploited for a long period of time, it is entirely possible that they may become endangered or even disappear altogether.
- Over-exploitation of natural resources compromises the availability and fair distribution of resources and their associated benefits to the health of ecosystems and the livelihoods and wellbeing of people.

1.2. Soils

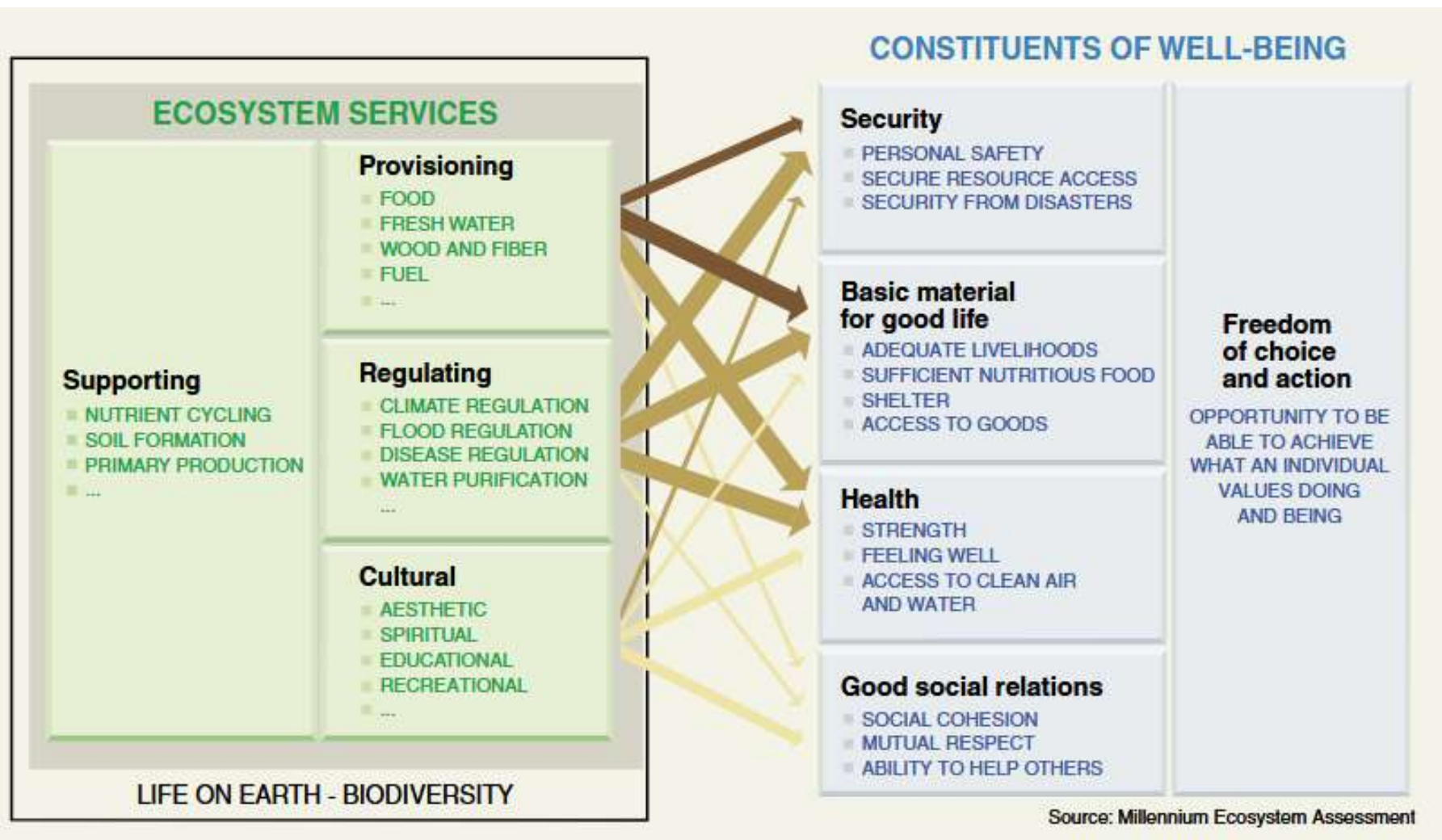


- Soils are fundamental to life on earth, and the ability of soils to provide safe and nutritious food is a key contribution to us and to nature as a whole.
- Soils are the foundation of our food security, providing 95% of the food we eat.
- Food starts with soils. About 95 percent of our food nutrients come from soils, which have a natural capacity to provide nutrients to support crop growth.
- Soil underpins the whole production system.

...1.2. Soils

- Probably, the most recent and appropriate definition of soil is:
A generally loose mineral and/or organic material at or near a planetary surface and usually holds liquids, gases, and biota and support plants.
- Soil forms so slowly that we should regard it as a non-renewable resource.
- Healthy soil , full of biological life, implies that all aspects of the soil, chemical, physical and biological are in a perfect status to be able to produce a healthy food. All people depend on nature and ecosystem services to provide the conditions for a decent, healthy, and secure life (Figure 1). Soil functions are at the core of virtually all processes. **Soil Ecosystem services:** Soil provides, regulates and supports a number of ecosystem services including cultural values.
- Soils are in danger of being the ‘orphan asset’, forgotten when food security or forests are discussed. The global decline in soil quality is therefore a major concern.

Figure 1. Linkages between Ecosystem Services and Human Well-being



...1.2.1. Major Components of Soil

Therefore, mineral soils consist of four major components. These are:

- Mineral (inorganic) constituent: extremely variable in size ranging from 0.0002 to 2 mm in diameter.
- Organic matter: represents an accumulation of partially decayed and partially synthesized plant and animal residues.
- Soil water: makes up soil solution, together with its dissolved salts, which is important as a medium for supplying nutrients to growing plants.
- Soil air: The amount and composition of soil air is determined to a large degree by the water content of the soil.

... 1.2.1. **Major** Components of Soil

Figure 2. Volume composition of a favorable /an ideal/ soil for plant growth



... 1.2.2. Soil formation

Soils are formed from rocks by weathering. The major parent material for soils are rocks. Soil formation takes place after a gradual exposure of rocks to the elements and activity of organism – called weathering.

Weathering includes

- physical weathering/disintegration - breaking or grinding particles to smaller sizes without changing their chemical composition.
- chemical weathering/decomposition - the dissolving or chemical alteration of minerals

1.2.2. Soil formation



Many types of plants have adapted to grow on rocks or new soil.

1.2.2.1. Agents

in

Physical Weathering

- Temperature fluctuations
- water/ice
- wind,
- gravity
- mechanical actions of animals, and
- mechanical forces of growing plants

Chemical Weathering

- **Oxygen**: in oxidation-reduction reactions
- **water**: in hydrolysis, hydration and (dis)solution
- **carbon dioxide**: in carbonation
- **organic and inorganic acids**
- **bases**, and
- **salts**

1.2.2.2. Processes in Soil Chemical Weathering

Hydrolysis: is the reaction between the H^+ and/or OH^- ions of water and the elements or ions of a mineral.

Oxidation-reduction reactions: Another chemical process involves the simultaneous loss and gain of electrons in substances.

Hydration: is the binding of water molecules to a mineral.

Carbonation The majority of soil organisms release carbon dioxide (CO_2) from respiration. Carbonation is a reaction by which water dissolves carbon dioxide and the dissolved carbon dioxide forms carbonic acid.

1.2.2.3. Factors of Soil Formation/Soil forming factors

The five soil-forming factors that influence the development of soil are:

- Parent Material
- Topography or position in the landscape
- Climate
- Living organisms, especially vegetation
- Time

1.3. Soils and Plants

In agriculture, using manure/fertilizer returns some organic matter to the soil. Any plant, even annual or perennial weeds, drop leaves and twigs into the soil, which becomes food for soil biota. At the same time, their shade reduces solar radiation from killing microorganisms. Therefore, they play a key role in improving soil fertility and making soil conducive for soil biota to thrive.

1.3.1. Essential Plant Nutrients

The criteria to consider an element to be essential are:

- A deficiency of an essential nutrient makes it impossible for the plant to complete the vegetative or reproductive stage of its life cycle.
- Such deficiency is specific to the element in question and can be prevented or corrected only by supplying this element.
- The element is involved directly in the metabolism of the plant.

...1.3.1. Essential Plant Nutrients

- Sixteen elements are known to be essential in plant growth. They can be grouped based on the amount they are required by plants and based on their sources. Eighteen nutrients are necessary for human health.

Macronutrients: nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), and sulfur (S).

Micronutrients: boron (B), zinc (Zn), manganese (Mn), iron (Fe), copper (Cu), molybdenum (Mo), chlorine (Cl).

- Though the majority of the elements come from the soil, they make up very small part of the plant tissue as plant tissues are mainly made of C, H and O (94-99.5%). C, H and O are not limiting under normal condition as they are supplied by the atmosphere and H₂O.

...1.3.1. Essential Plant Nutrients

A healthy soil is a chemically balanced soil. The macronutrient essential soil nutrients, very important for plant growth, are.

1.3.1.1. Primary Nutrients

🌿 Nitrogen is the major nutrient for strong, vigorous growth, dark green leaf color, and photosynthesis.

🌱 Phosphorus helps roots for strong root growth and development.

🌸 Potassium helps flowers and fruits to develop, improving the overall health and strength of the plant.

1.3.1.2. Secondary Nutrients

Calcium is important for general plant vigor.

Magnesium helps regulate the uptake of other plant foods.

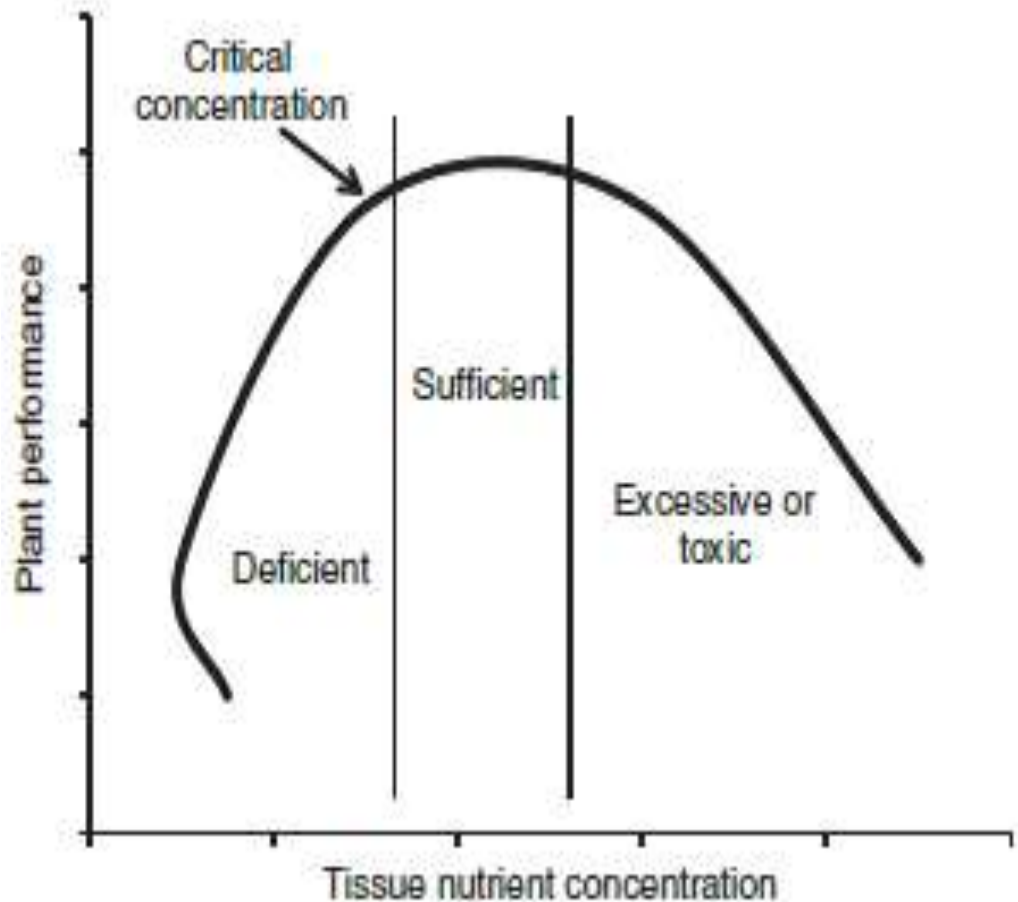
Sulfur helps maintain a dark green color while encouraging more vigorous plant growth.

...1.3.1. Essential Plant Nutrients

- Five additional elements – sodium (Na), cobalt (Co), vanadium (Va), nickel (Ni) and silicon (Si) are essential elements for some plants. Plants absorb many non-essential elements. The content of the micronutrients is also often far in excess of physiological requirements.
- Nutrients are desirable in particular concentrations and can be detrimental to plant growth when concentrations in soil solution are either too low (limiting) or too high (toxicity) [Figure 3].
- The lack of essential nutrients, including macro and micronutrients, leads to the underdevelopment of plants and a decrease in yield and in crop nutritional value.

Nutrient Uptake by Plants

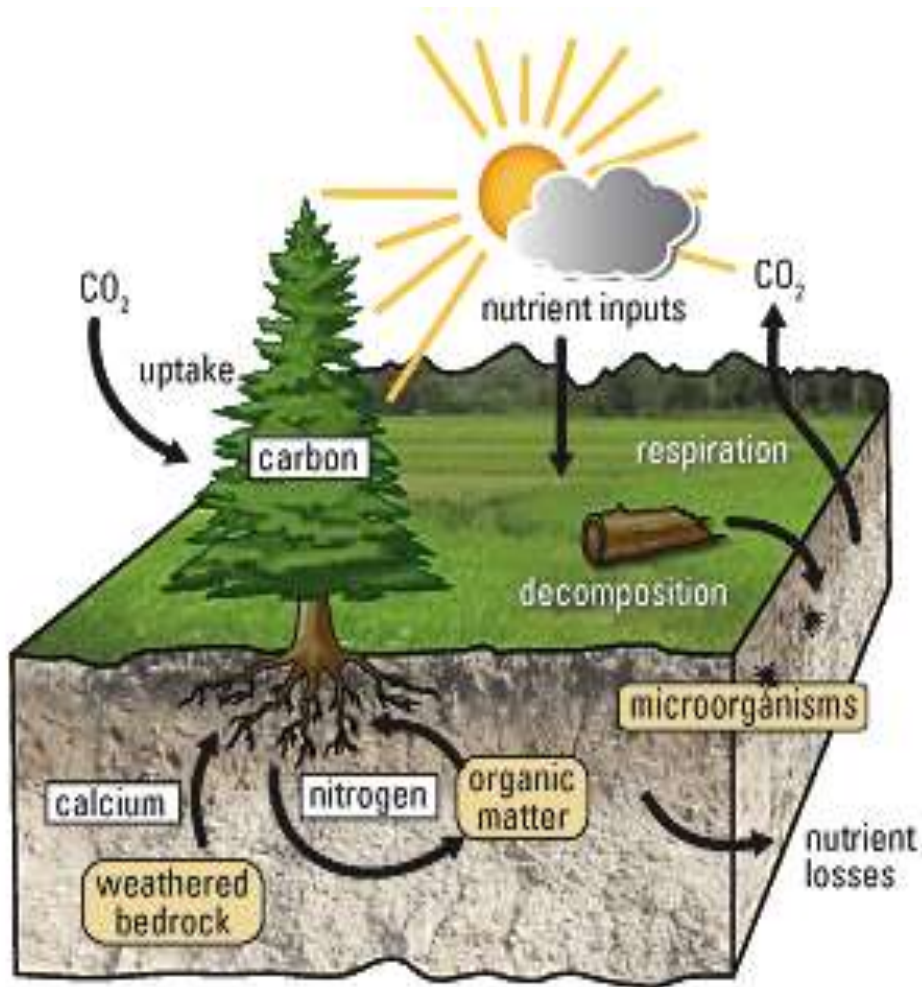
Figure 3. Nutrient concentration in plant tissues as it relates to plant performance. Below the critical range plants are considered deficient. The range where nutrient concentration in the tissue increases, but performance does not, is considered the sufficient range. At some point beyond the sufficiency range, excessive nutrient concentrations can limit plant performance and this is considered the excessive or toxic range.



1.3.2. Soil-Plant Relations

- Beneficial aspects of soils to plants include providing physical support, heat, water, nutrients, and oxygen. Heat, light, and oxygen are also obtained by the atmosphere, but the roots of many plants also require oxygen. Nutrients, dissolved in soil water solution, are derived from soil minerals (inorganic) and organic material (see Figure 3).
- Though many aspects of soil are beneficial to plants, excessively high levels of trace metals (either naturally occurring or anthropogenically added) or applied herbicides can be toxic to some plants.
- The ratio of solids/water/air in soil is critically important to plants for proper oxygen levels and water availability. Too much porosity with air space, such as in sandy or gravelly soils, can lead to less available water to plants, especially during dry seasons when the water table is low. Too much water, in poorly drained regions, can lead to anoxic conditions in the soil, which may be toxic to almost all plants.

Figure 4. Soil-Plant Nutrient Cycle



1.4. Key Pillars of Resource Management

The **main parts** of natural resource management are:

- (1) Forest management/(Agro)forestry
- (2) Water resource management i.e. irrigation and water conservation/harvesting
- (3) Soil resource management and
- (4) Agriculture resource management

Forestry – which yields a vast range of products and ecosystem services useful both for local communities and at a global scale – could play a key role in accelerating a transformation towards societies that simultaneously conserve nature, better provide for human well-being, and generate income, particularly for rural people.

Forestry products can be:

- wood/timber and
- non-timber /non-wood/ forest products (NTFPs/NWFPs)]. Forests create clean air, fertile soils and rich biodiversity.

Agroforestry, an ecologically based traditional farming practice, is the use of trees and shrubs as part of agricultural systems. Planting a range of trees, perennial plants and crops to ensure vegetation stays in the ground year-round.

Sustainable management of forest and agroecosystems requires the maintenance of the supporting natural resources and ecosystem services.

Water resource management includes water supply management, excess water management and ecosystem restoration.

1.5. NR Degradation & Conservation

1.5.1. NR /Environmental/ Degradation

- Land degradation is the progressive reduction in the productive potential of the land to sustain life and provide food security. It is a natural and human-induced phenomenon.
- The world's land area is now 40% degraded and accelerating. This means that greater risk to the climate, biodiversity and the world's ability to feed a growing population. Land is considered 'degraded' when most of its natural resources, soil, water, biodiversity, trees or native vegetation is depleted, many of them which include "green" areas that were intensely farmed or deforested.
- It can be classified under four headings as:
 - 1) Soil degradation
 - 2) desertification
 - 3) vegetation cover degradation
 - 4) water resources degradation

1.5.1.1. Types of Soil Degradation

- Soil degradation is the process which lowers the current and/or future capacity of the soils to produce goods or services. It includes:
 1. Soil erosion: is the displacement of soil from one place to the other. Topsoil is particularly affected and fertility losses are high in relation to physical losses.
 - Water erosion: This is the most widespread soil degradation type in SSA.
 - Wind erosion: Wind erosion occurs most frequently in the arid and semi-arid parts of SSA. If the climate becomes drier or windier, wind erosion is likely to increase.

... 1.5.1.1. Types of Soil Degradation

2. Soil chemical degradation includes processes such as salinity /salinization/ and sodification, acidification, nutrient mismanagement, leaching and soil pollution /build-up of toxicity/.

Agricultural systems lose nutrients with each harvest, and if soils are not fertilized, fertility is progressively lost, and soils will produce nutrient-deficient plants. Nutrient depletion (gradual loss of fertility) occurs if agriculture is practiced on poor or moderately fertile soils, without sufficient application of organic or chemical fertilizers.

Nutrient imbalance is a significant obstacle on the road to food security as it directly affects food production, quality and nutritional value, and safety. Soil nutrient loss is a major soil degradation process threatening nutrition and is recognized as being among the most important problems at a global level for food security and sustainability.

3. Soil physical degradation includes crusting, compaction, soil sealing and waterlogging, etc.
4. Soil biological degradation includes loss of humus, loss of soil biodiversity or lowering of soil fauna/flora (esp. lowering of the population, activity and diversity of soil micro-organisms).

1.5.1.2. Causes of Land Degradation

World soil health is under pressure from several threats including erosion, loss of soil organic carbon and biodiversity, pollution, and salinisation. Soil nutrient loss is a major soil degradation process threatening nutrition and is recognized as being among the most important problems at a global level for food security and sustainability.

The global pressures on agricultural lands are often anthropogenic in origin, for example, climate change, erosion, or land-use change (FAO, 2015; Smith et al., 2016). Pressures, for example, deforestation, overfishing and water scarcity are all working in the same direction. They are undermining and degrading nature.

- Deforestation and removal of natural vegetation and crop residues
- Overgrazing
- Inappropriate agricultural practices /exploitative use of the land/ such as:
 - Cultivation of marginal lands (steep slopes, shallow soils, etc)
 - Overcultivation (excessive tillage) or continuous cultivation
 - pesticides and fertilizer overuse
- (Bio-)Industrial activities and mining
- Urbanisation and (unsustainable) infrastructure

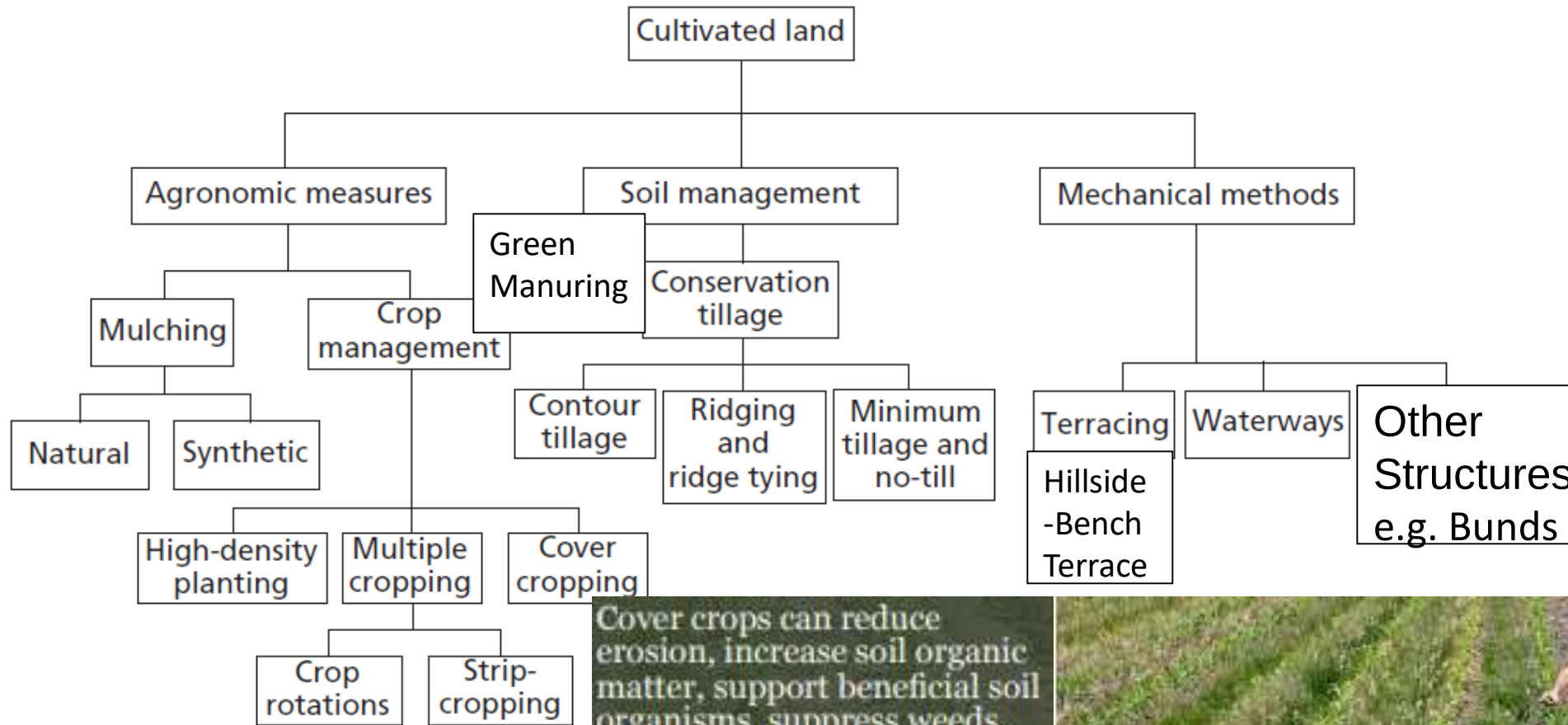
1.5.2. Environment/NRs Conservation

1.5.2.1. Soil and water conservation

- In hilly or mountainous areas, soil erosion is unavoidable due to rain and wind. Soil conservation is becoming increasingly important due to climate change and harmful farming practices.
- Soil and water conservation are those activities which maintain or enhance the productive capacity of the land including soil, water and vegetation in areas prone to degradation through
 - prevention or reduction of soil erosion,
 - conservation or drainage of water and
 - maintenance or improvement of soil fertility. Simply increasing soil organic matter alone improves moisture holding capacity and soil structure, so when it rains, more of it soaks into the ground, reducing flooding.
- Soil conservation practice includes soil management, crop management, engineering, range management and forestry operations.
- Terracing is one of the engineering practices used to prevent rainfall runoff in sloping lands for soil and water conservation.

...SOIL AND WATER CONSERVATION MEASURES

Fig 5. Soil conservation strategies for cultivated land



Cover crops can reduce erosion, increase soil organic matter, support beneficial soil organisms, suppress weeds, and release chemicals that are toxic to soilborne pathogens.

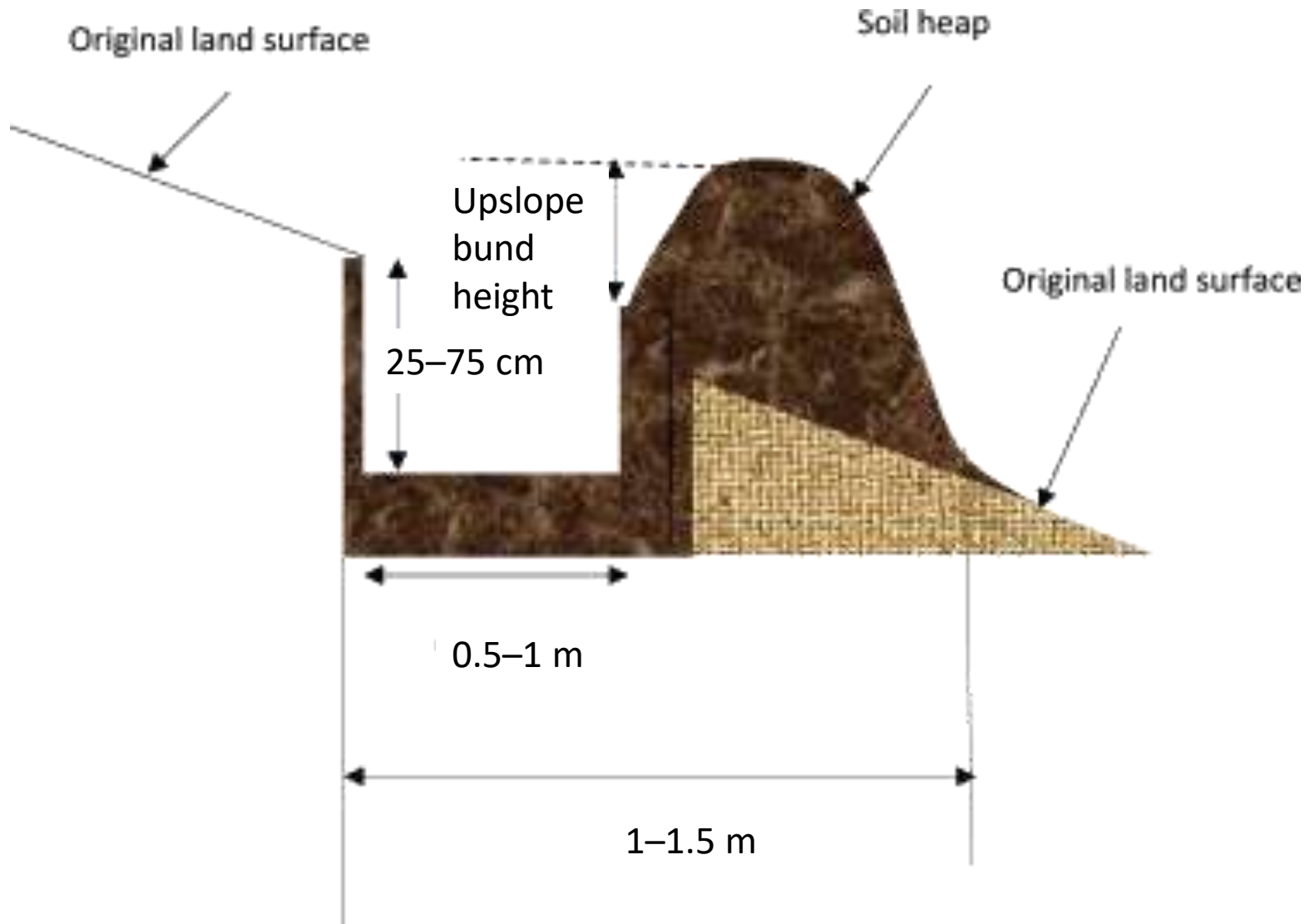
Cover crops also enhance soil biodiversity.

Multispecies cover cropping better manages soil health and improve sustainability.



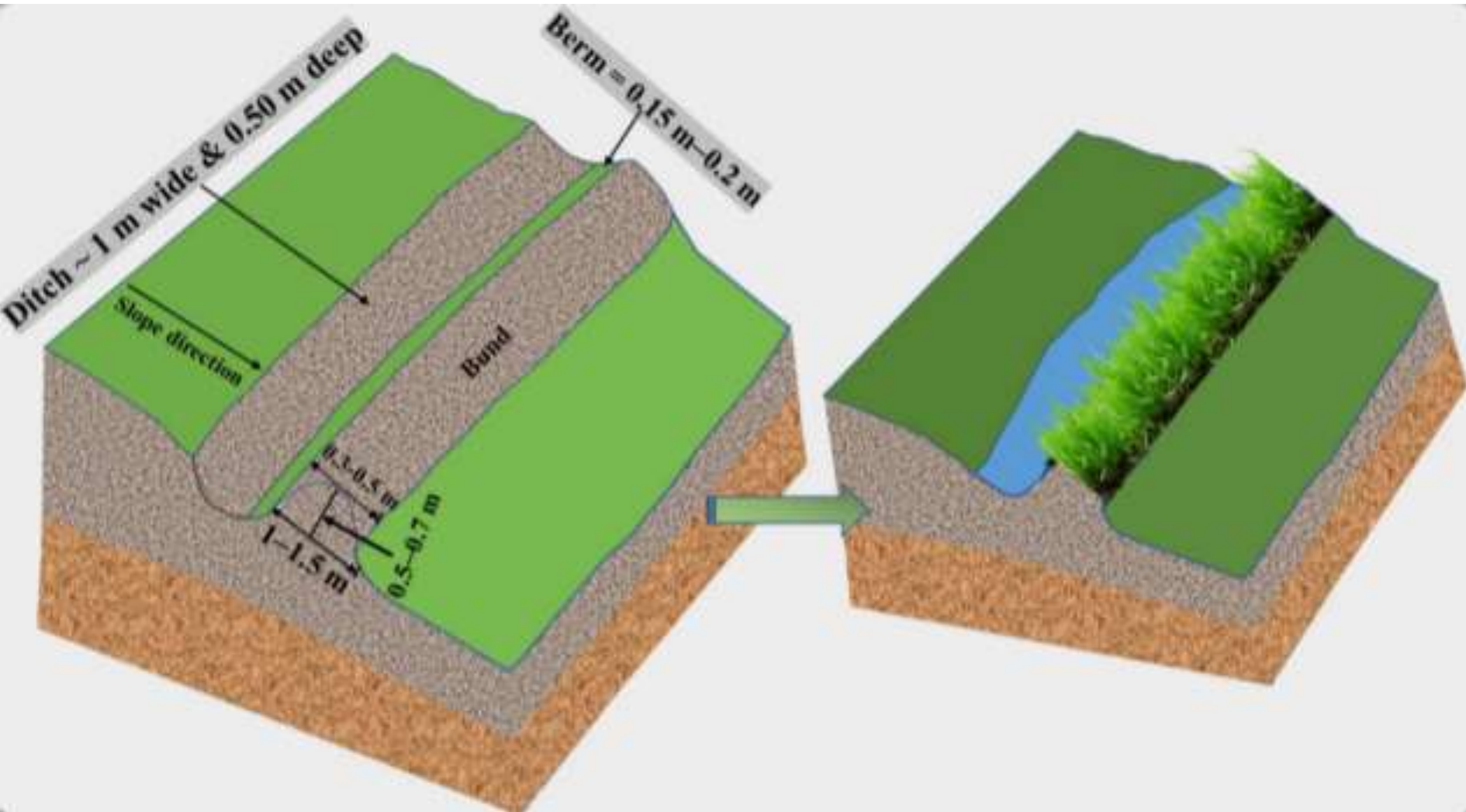
...SOIL AND WATER CONSERVATION MEASURES

Fig 6a. Side view and dimension of a typical soil bund



...SOIL AND WATER CONSERVATION MEASURES

Fig 6b. Side top view and dimension of a typical soil bund of soil bunds before (left) and after (right) being reinforced by grass or legumes



... 1.5.2.1. Soil and water conservation

2. CONSERVATION MEASURES ON GRASSLAND

- Controlled Grazing
- Zero-grazing: *Cut and Carry (stall-feeding)*
- Grassland Improvement



3. CONSERVATION MEASURES ON FORESTLAND

- Hillside Terrace
- Tree Planting

⋮

4. CONSERVATION MEASURES COMMON TO ALL LAND USE TYPES

- Area Closure
- Cutoff Drain
- Revegetation
- Water Harvesting
- Waterway
- Trash line

⋮

1.5.2.2. Impact of NR conservation to food, post-harvest and nutrition

1. Role of (Agro)forestry Practices for nutrition

- In association with food and agriculture, NWFPs/NTFPs provision includes
 - ✓ foods (nuts, fruits, mushrooms, honey, game, gums);
 - ✓ food additives (spices, herbs, flavorings, sweeteners); and
 - ✓ Fodder
- Agroforestry is also one of the best uses of agro-biodiversity that generates multiple benefits, i.e. ameliorates soil fertility, controls erosion and enhances moisture retention which
 - improves productivity and production of diversified food products and income for households and communities. E.g. Alley cropping is an agroforestry system in which food crops are grown in alleys between rows of hedges and it can be used as SWC practice on cultivated land.
- Inclusion of nutritionally rich/dense and diversified plants in the agroforestry system is essential to address malnutrition. This can also support the future food needs of the growing population through
 - ensuring people to have access to adequate and
 - nutritious food produced in an environmentally and socio-culturally sustainable manner.

... 1.5.2.2. Impact of NR conservation to food, post-harvest and nutrition

2. Link between soil management and nutrition

- Any thriving natural or agricultural ecosystem begins with soil. Thus, soil health is the starting point for the transformation of the entire food chain/web. Healthy and nutritious food for crops, livestock, and human starts with healthy soils.
- Improving soils can boost crop production by 40%. A balanced supply of soil nutrients is essential to support crop and forage grass yields and the nutritional value of the agri-food system. The lack of basic nutrients, i.e. macro and micronutrients, leads to the underdevelopment of plants and decrease in yields and crop nutritional value. Soil macro- and micronutrient deficiencies limit plant health, crop productivity and the nutritional quality of food, which together ultimately affect nutrition and human health.
- Soil fertility has a great role in producing sufficient, safe, and more nourishing food for healthier plants, animals, and people. Improved soil management yields a fertile (including high nutrient content) soil which in turn is mirrored in the crops/plants. Soil fertility and nutrition involve processes at scales ranging from molecules to the entire planet. Generally, soils offer a more nutritious agrifood system for enhancing human health and wellbeing while protecting the environment.

... 1.5.2.2. Impact of NR conservation to food, post-harvest and nutrition ...

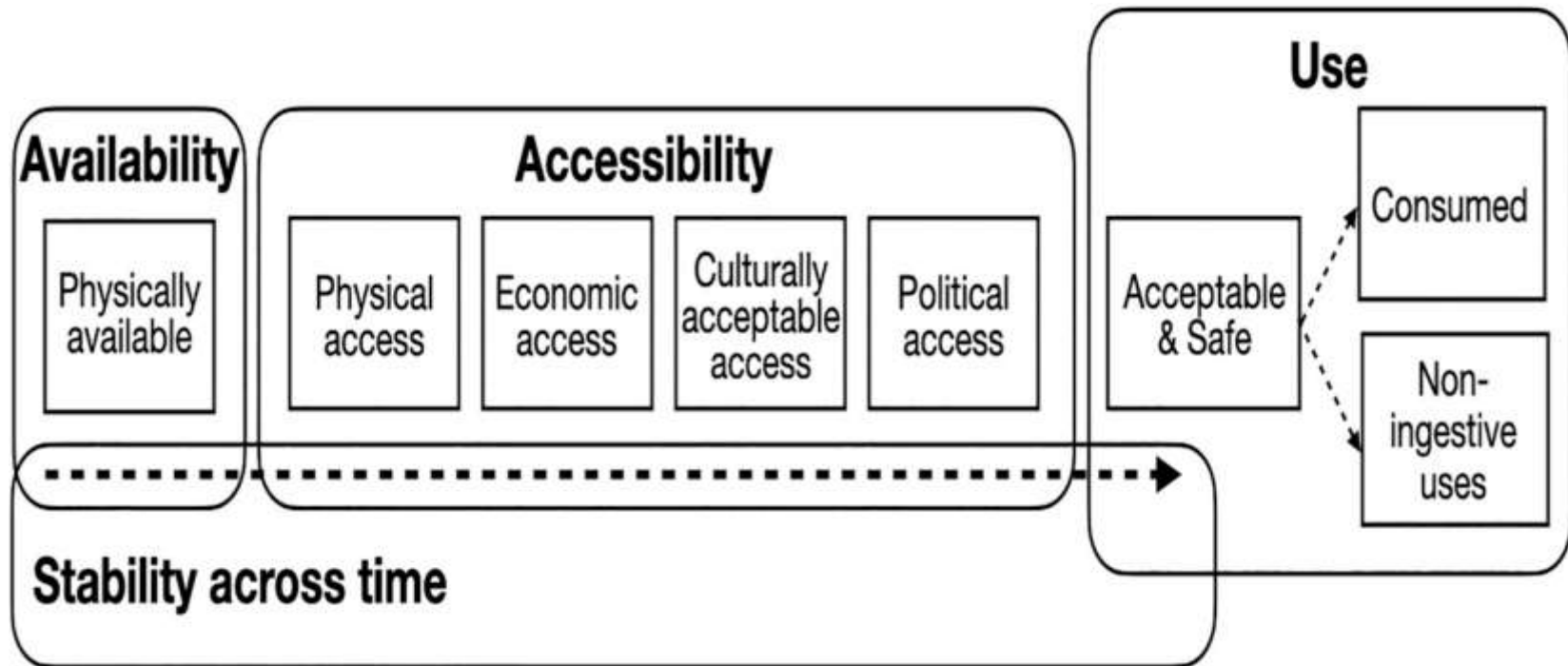
3. Irrigation and water conservation/harvesting for nutrition

- Irrigation increases food production and availability. Irrigation schemes are likely to provide enhanced food security and nutrition to farmers due to a greater availability and stability of food supplies and crop diversification.
- The use of irrigated agriculture enables crop production, often focused on high-value crops, during the dry season; specifically, it increases the number of harvests to 2-3 per year and leads to increased yields and crop diversification. This contributes to the improvement of nutrition.
- Smallholder irrigation systems are mostly used to grow vegetables in the dry season;
 - consequently, vegetable consumption among irrigation users and their communities usually increases.
 - Irrigation and water harvesting may also provide the water requirement of fruit seedlings all-year-round.
 - Vegetables and fruits are rich in micronutrients and provide important benefits, especially for children.
- Water scarcity is one of the most pressing challenges of our time. As the largest water user worldwide, the agriculture sector is uniquely positioned to drive the transition towards sustainable water management.

... 1.5.2.2. Impact of NR conservation to food, post-harvest and nutrition ...

... 3. Water Security for Food Security, Good Nutrition, and Well-being

- The 4 domains of water security are of importance for humanity are its availability, access, stability, and quality.



- Consumption-related water use includes water that is drunk as well as eaten. Water is found in all foods, and some cooking techniques (e.g., boiling, steaming) substantially increase their water content. The noningestive uses of water are far more numerous. These include practices for sanitation and hygiene—for example, washing hands, dishes, and foods; bathing; laundry; and sewage disposal.

...1.5.2.2. Impact of NR conservation to food, post-harvest and nutrition

4. Watershed management and nutrition

- Watershed is an area of land that drains water to a common outlet.
- Through the creation of ecologically clean (eco-clean) watersheds, the soil and water loss control and economic development can be achieved in a coordinated way to better benefit the local people.
- The pattern/procedure for the creation of eco-clean watersheds is governing a clean small watershed, revitalizing a village, creating a demonstration project, cultivating a green industry, and making people rich.
- The eco-clean watershed management modes are **protecting water** via
 1. controlling hilly area - control pollution by reducing sources, intercept sewage and purify water – ecological restoration and
 2. protecting hilly area - protect water via controlling slope – purify water flowing to villages – maintain clean water quality while developing (Fig. 7).

...1.5.2.2. Impact of NR conservation to food, post-harvest and nutrition

...4. Watershed management and nutrition

Fig 8. Ecological clean (eco-clean) watershed management modes



...1.5.2.2. Impact of NR conservation to food, post-harvest and nutrition

...4. Watershed management and nutrition

- Sound management of watersheds provides multiple benefits in terms of food production, environmental health, and nutrition.
- Properly managed watersheds (rangelands, forests, etc) enhance ecosystem services that boost agricultural productivity, increase access to clean water, replenish aquifers, mitigate damage from storms and floods, and reduce negative environmental impacts from agriculture.
- Therefore, developing and properly managing watershed could contribute to the improved nutritional status of smallholder farmers through
 - enhancing agricultural production and diversification.

...1.5.2.2. Impact of NR conservation to food, post-harvest and nutrition

Significance of NR conservation to food and nutrition

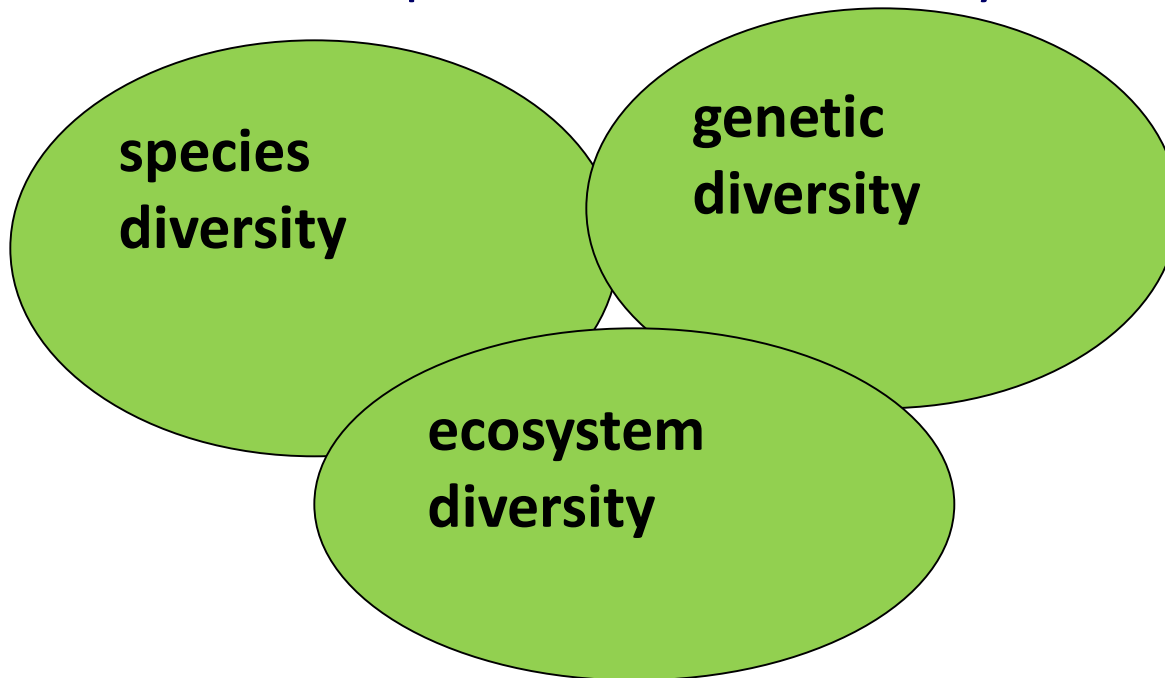
NR conservation e.g. By soil health, we can improve water quality and water quantity, and ultimately support a healthier planet. Moreover, the access and use of water affects many other nutrition-related phenomena, such as agricultural productivity, food-preparation practices, eating behaviors, dietary diversity, infant and young child feeding practices, and energy expenditure.

Significance of NR conservation to food and post-harvest

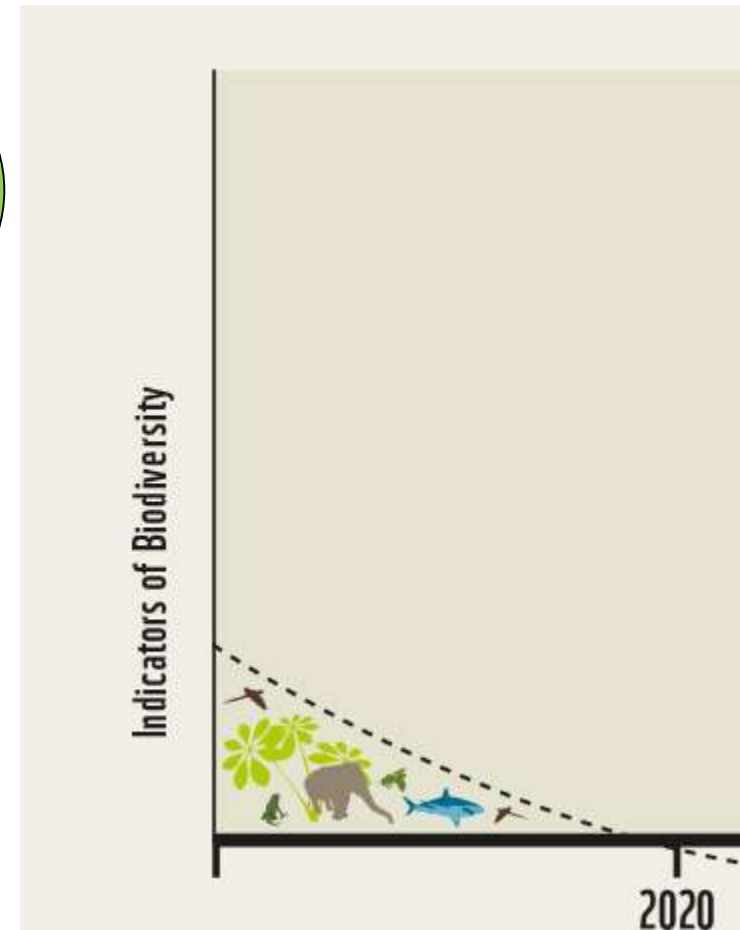
- The urgent challenge that the agricultural field is facing is to continue feeding the world's constantly growing population as resource availability steadily declines. Resources are used for producing food that will either never reach the consumer or will never get consumed.
- Farmers in a developing country lack the capital to invest in technologies, such as proper packing material and cooling systems.
- Natural resources conservation helps to decrease this food loss and post-harvest in two ways.
 1. By using resources more efficiently
 2. Recycling as much as possible OR turning waste into wealth/treasure

1.5.3. Biodiversity and Nutrition

- Agricultural biodiversity is important for food and nutritional security.
- The components of biodiversity are



A species range is an area where a particular species can be found during its lifetime. Species ranges include areas where individuals or communities may migrate or hibernate. Several factors determine species range; Climate is one important factor.



... 1.5.3. Biodiversity and Nutrition...

- Agriculture takes up space. Prime farmland — land with good soil and water access — is a limited resource. These same areas often support rich wild ecosystems like prairies and forests; converting these areas into farms eliminates much of that wild biodiversity. Unfortunately, agriculture's continually expanding footprint places these sensitive and important wild areas at risk of destruction. This process of bringing more wild land into agriculture (to produce more food) is called extensification.
- Monocultures may offer consistency to farmers and the food industry, but they pose a risk to food security.
- Expansive agriculture is responsible for so much habitat loss, preventing wild lands from being converted into farmland is critical to maintaining biodiversity.
- Wild populations are a store of potentially useful genes that have been bred out of their cultivated relatives.
- Nature-positive development can only be achieved through transformative change, addressing the underlying societal factors that are the root causes of biodiversity loss and that drive development. A whole-of-society approach (to Biodiversity Governance) recognizes and builds on a society-wide mobilization of actors, ranging from non-state actors (such as indigenous peoples and local communities, NGOs, and the private sector) to all levels of government.

... 1.5.3. Biodiversity and Nutrition



Biodiversity gives
the **food we eat**,



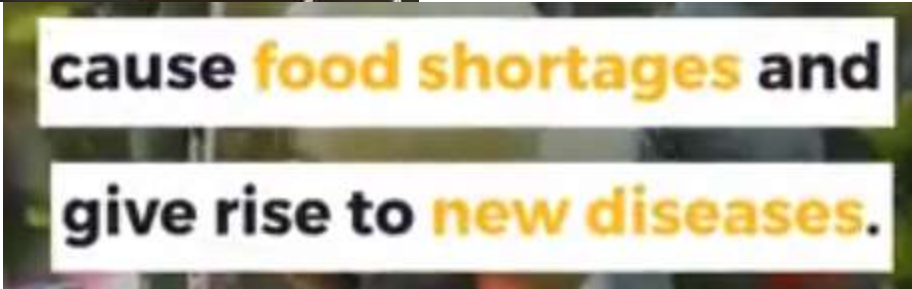
air to breathe and
water to drink.



80% of biodiversity on
land is in forests



The loss of biodiversity will
raise global temperatures,



cause **food shortages** and
give rise to new diseases.



Scientists say that around 27000
species of life forms are becoming
extinct every year due to loss of
habitat.



Loss of biodiversity further disrupts the
soil habitat and prevents soil
regeneration.

... 1.5.3. Biodiversity and Nutrition...

- The biodiversity contributes to agriculture and food production in one way or another, is crucial to food security, nutrition and health, sustainable development and the supply of many vital ecosystem services. Biodiversity and healthy ecosystems are critical to food systems. Biodiversity includes a wide diversity of genetic resources which underpin vibrant food systems.
- Better conservation of biodiversity will have positive consequences on reducing malnutrition , and thus, poverty. Pollinators affect crop yields, thus market prices, and the nutritional value of foods. Pollinating animals like bees are so crucial for producing nutrient-rich crops that more than half of the population in some developing countries would experience malnutrition (nutrient deficiencies, especially of Vitamin A) if pollinators disappeared. The very diverse species that live in soils make up the often overlooked biodiversity that maintain healthy soils, helping micronutrients reach plants and ultimately our plates.
- ❑ The challenges of halting biodiversity loss and meeting future food requirements are extremely linked. So, halting biodiversity loss requires immediate action. Halting and reversing biodiversity loss requires a sharp increase in the area of natural ecosystems where biodiversity can thrive, through stronger conservation and restoration actions.
- Mainstreaming biodiversity in society and all economic sectors can sustainably address the issue of malnutrition. Embedding biodiversity conservation into land-use planning and food production systems is also an opportunity to achieve food and nutrition security (SDG 2) and meet the Paris Climate Agreement targets.

... 1.5.3. Biodiversity and Nutrition...

- **Biodiversity** conservation can be beyond classical on-farm conservation approaches (e.g. conservation of local crop varieties and crop wild relatives) and may focus instead on the contributions of agroecology to mitigate species decline and ecosystem degradation at the landscape scale. This although important is less explicitly recognized within existing narratives of agroecology or conservation communities.
- By embracing both traditional knowledge and new research, farmers and scientists are producing food in a way that harnesses biodiversity to make the most of what nature provides. This approach is called agroecology, and is a core component of regenerative agriculture, which builds up natural resources like healthy soil and water rather than using them up.
- The adoption of safe farming methods and the use of farmer-owned seeds traditionally used in the country are the only ways to ensure food security (a continuous supply of food for the population). The **typical** diversity of seed in an household seed bank should include several grains (including farmer varieties) of maize, sorghum and pearl millet; legumes, which are key to food security and soil fertility; traditional fruit varieties; and nuts. Peanuts can be used in place of oil.

4. CLIMATE CHANGE IN AGRiculture and INTERVENTIONS

- **Weather refers to short term atmospheric conditions while climate is the weather of a specific region averaged over a long period of time.** Climate change refers to long-term changes and can alter weather patterns and climate.
- Agriculture is a risky business that is frequently exposed to uncertainties and is still the most vulnerable sector to climate change.
- Climate change increases the probability of extreme events and catastrophic risks affecting agriculture.
- Climate crisis is projected to increase global temperatures by 1-2°C, from the pre-industrial level, by 2050. For every 1°C increase, the global yields of the three major cereals— rice, wheat, and maize— are expected to decrease by 3-7%.
- Africa is suffering more than any region to the effects of extreme weather events and climate change.
- While the effects of climate variability are being felt on the field, the vast majority of African farmers have little or no capacity, to deal with climate change on their own, as the impact of climate change has destroyed the social and economic sustainability of family farming and has aggravated the food security challenges of a rapidly growing population.

4.1. Introduction to Climate Change

- is modification of earth's climate brought about as a result of changes in the atmosphere as well as interactions between the atmosphere and various other geologic, chemical, biological, and geographic factors within the earth system.
- is attributed to human activity, altering the composition of the global atmosphere. How we invest and consume will influence whether we meet the climate challenge and whether threatened communities and ecosystems can adapt and survive.
- Human activity includes the pollution that arises from industrial activity and other sources that produce greenhouse gases (GHGs).
- These gases, such as carbon dioxide, have the ability to absorb the spectrum of infrared light and contribute to the global warming of our atmosphere.
- Once produced, these gases can remain trapped in the atmosphere for tens or hundreds of years.
- Climate change is already happening, and is projected to continue in the future.